HACKTOBERFEST HACK DAY GUELPH • GDG ON CAMPUS

MEDIQ: Executive Pitch & Submission Blueprint

Hacktoberfest Hack Day Guelph × GDG on Campus Guelph
Sunday, October 19, 2026 • Cutoff: 10:00 PM EDT

🩺 Executive Summary

MEDIQ is an air-gapped, touchless clinical workstation co-pilot engineered for surgical operating rooms, emergency departments, and cleanrooms. It resolves the two most dangerous failure modes in modern medical computing: **cross-contamination from touching physical keyboards/screens**, and **acoustic spoofing in ambient voice AI**.

Core Architecture:

1. **Zero-Trust On-Device Biometrics:** WeSpeaker ResNet34 ONNX running in browser WASM. Extracts 256-D speaker embeddings in under 40ms; enforces fail-closed acoustic verification with zero audio cloud leakage.
2. **Sterile 3D Air Gestures:** MediaPipe Vision running in WASM. Tracks 21 hand landmarks at 30+ FPS for pinch-dragging, two-hand zooming, and peace-scrolling without physical touch.
3. **Dual-Model Clinical Consensus:** Open-weight **Hermes 3** (DigitalOcean GenAI) synthesizes doctor dictations into structured SOAP notes; **Google Gemini 2.0 / Flash** independently audits drafts against multi-layered EHR history to catch lethal contraindications (e.g. Penicillin / Amoxicillin anaphylaxis).
4. **Air-Gapped Deterministic Safety:** Hardcoded clinical rules run locally if network or cloud connectivity drops.

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1. The 3-Minute Live Stage Pitch & Demo Choreography

Elapsed Time	Presentation Stage	Spoken Pitch (Word-for-Word)	On-Screen Demonstration
0:00 – 0:35	**The Hook: Sterile Field Crisis**	*"In operating rooms and ICUs, surgeons face a dangerous dilemma: to look up a CT scan or check an allergy, they must either break scrub and touch contaminated keyboards—or yell to an unverified assistant. Meanwhile, ambient hospital voice AIs have a fatal flaw: they fail open. Anyone walking into the room—a visitor or an orderly—can trigger record changes.* *We built **MEDIQ**: the first touchless clinical workstation co-pilot powered by on-device biometric defense and dual-model AI consensus."*	**Screen:** Open `http://localhost:5173`. Point out top bar indicator: `Voice Biometrics: Armed (Fail-Closed)`.
0:35 – 1:10	**The 3-Layer Architecture**	*"MEDIQ operates entirely hands-free across three clinical defense layers:* *1. **Zero-Trust Edge Biometrics:** On-device WeSpeaker ResNet34 ONNX running in browser WASM. It calculates acoustic embeddings in under 40ms. Raw audio is discarded immediately.* *2. **Touchless Sterile Computer Vision:** MediaPipe 3D hand tracking in sterile air.* *3. **Dual-Model Consensus Engine:** Open-weight **Hermes 3** on DigitalOcean GenAI synthesizes clinical notes, while **Google Gemini** independently audits against EHR history."*	**Screen:** Highlight Doctor Enrollment card and AI Settings showing Hermes 3 + Google Gemini consensus.

Elapsed Time	Presentation Stage	Spoken Pitch (Word-for-Word)	On-Screen Demonstration
1:10 – 1:35	**Demo 1: Biometric Intrusion Rejection**	*"Watch what happens when an unauthorized voice attempts access: 'Discharge patient 104'.* *Access Denied. Cosine similarity drops below our 0.50 margin. The system fails closed, logs the attempt to the clinical audit trail, and executes nothing."*	**Action:** Click **Demo Quick-Sim ➔ Simulate Unknown Speaker**. Glowing red warning appears: `Voice Not Verified: Access Denied`. Show Activity Log.
1:35 – 2:00	**Demo 2: Verified Doctor & Sterile Gestures**	*"Now the enrolled clinician speaks: 'Open patient 104'.* *Instant verification in 40ms. Now, the surgeon's hands are sterile and gloved. In mid-air: a peace sign scrolls patient vitals. A sterile pinch drags the lab report, and a two-hand outward gesture resizes the CT summary in mid-air—zero glass touched."*	**Action:** Click **Demo Quick-Sim ➔ Simulate Authorized Doctor** (or speak live). Patient 104 snaps open. Demonstrate floating report drag/resize.
2:00 – 2:35	**Demo 3: The Climax — Catching a Lethal Allergy**	*"Here is MEDIQ's crowning safety feature. A physician dictates: 'Patient has acute pharyngitis. Prescribe Amoxicillin 500mg TID for 10 days.' Watch what happens when we draft this note.* *Look at this red alert:* ***CRITICAL CONTRAINDICATION: Documented severe penicillin allergy with anaphylaxis.*** *Google Gemini cross-referenced the active prescription draft against EHR history, catching a cross-reactive aminopenicillin before it*	**Action:** Open **Visit Notes ➔ Demo Quick-Sim ➔ Simulate Clinical Dictation** (Amoxicillin). Click **Generate structured draft**. Giant glowing red alert banner appears with Penicillin contraindication badge!

Elapsed Time	Presentation Stage	Spoken Pitch (Word-for-Word)	On-Screen Demonstration
		ever reached the pharmacy. Even if the internet cuts out, MEDIQ's deterministic offline rule engine guarantees safety alerts never fail."	
2:35 – 3:00	**The Winning Close**	*"MEDIQ is 100% open-source under MIT, requiring zero specialized medical hardware—it runs on any hospital laptop in a single* `docker compose up` *command. By pairing DigitalOcean's open-weight AI with Google Gemini's safety auditing and on-device WASM biometrics, we eliminate cross-contamination and prevent fatal medical errors. Thank you."*	**Action:** Show terminal running Docker container and repo badges. Open for Q&A.

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2. 60-Second Video Demo Storyboard (For DevRelay Link)

- **[0:00 – 0:10] Hook:** Hands raised in surgical posture. *"Touching a clinical workstation breaks sterility. Ambient voice AI risks acoustic spoofing. Meet MEDIQ."*
- **[0:10 – 0:25] Biometrics in Action:** Split screen showing audio capture. Trigger **Simulate Unknown Speaker** (`Access Denied`), then speak *"Open patient 104"* (`Verified · 40ms`).
- **[0:25 – 0:42] The Lethal Intercept:** Show clinical dictation proposing `Amoxicillin 500 mg` . Click *Generate structured draft*. Zoom into the vibrant red warning banner: **Google Gemini / Deterministic Safety Catch — Anaphylaxis Penicillin Allergy**.
- **[0:42 – 0:52] Sterile Gestures:** Camera overlay showing 21 hand landmarks. Pinch-drag floating lab report, two-hand zoom, fist to dismiss.
- **[0:52 – 1:00] Tech Stack Close:** Flash of `docker compose up` , DigitalOcean Hermes 3, and Google Gemini API badges. *"Open-source, fail-closed, lives saved."*

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3. Official DevRelay / Hackathon Submission Copy

Project Name

`MEDIQ — Medical Intelligence & Voice-Biometric Touchless Co-Pilot`

Source Code URL

`https://github.com/GurkiratA/Mediq.git`

Prize Categories Selected

- [x] **DigitalOcean: Best Open-Source AI Project**
- [x] **GDG on Campus: Best Use of Google Gemini API ($50 Bounty)**

AI Technologies Used Checklist

- [x] **Gemini** (Google Gemini 2.0 / Flash API)
- [x] **Claude** (Claude Code test harness)
- [x] **Open-Source / Open-Weight AI** (Hermes 3 via DigitalOcean GenAI + WeSpeaker ResNet34 ONNX via WASM)

Submission Description (Markdown)

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# MEDIQ – Medical Intelligence & Voice-Biometric Touchless Co-Pilot

> **Fail-closed voice biometrics, sterile 3D computer vision gestures, and dual-model
> *Built with Hermes 3 on DigitalOcean GenAI, Google Gemini 2.0, ONNX Runtime WASM, M

### 💡 Inspiration
In operating rooms, ICUs, and emergency bays, medical personnel face two major failur
1. **The Sterility Dilemma:** Consulting an EHR or zooming diagnostic images requires
2. **The Voice Spoofing Dilemma:** Ambient hospital voice assistants typically **fail

We engineered **MEDIQ**: an entirely touchless workstation co-pilot that strictly ref

### 🔬 What It Does
- **Fail-Closed On-Device Voice Biometrics:** Spoken commands (*"Open patient 104"*,
- **Sterile 3D Hand Gestures:** Powered by MediaPipe Vision in WASM. Clinicians pinch
- **Dual-Model Clinical Consensus (Hermes 3 + Google Gemini):**
  - **Drafting Engine:** Open-weight **Hermes 3** (hosted on DigitalOcean GenAI) synt
  - **Safety Auditing Engine:** **Google Gemini 2.0 / Flash** acts as an independent
- **Lethal Allergy Intercept:** Proactively detects cross-reactive contraindications.
- **Deterministic Air-Gapped Fallback:** If internet access drops, deterministic offl

### ⚙️ How We Built It
- **Biometric Edge:** WeSpeaker ResNet34 ONNX inside `onnxruntime-web` (single-thread
- **Vision Layer:** `@mediapipe/tasks-vision` tracking 21 3D hand landmarks in real t
- **Backend & AI Orchestration:** Express 5 + Node.js 22 with Zod contract validation
- **Frontend Workstation:** React 18 SPA with Tailwind CSS v4, custom 1€ filtering fo
- **Deployment:** Fully dockerized multi-stage build running locally in a single comm

### 🏆 Prize Category Alignment
- **DigitalOcean Best Open-Source AI Project:** 100% MIT-licensed, powered by open-we
- **GDG Best Use of Google Gemini API ($50 Bounty):** Deploys Gemini in a high-stakes

### 🛡️ Rigorous Verification
- **154 Unit Tests Passed** across 14 Vitest suites (deterministic safety engines, ce
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- **End-to-End Browser Tests:** Full Playwright suite validating doctor enrollment, v
- **Zero Leaked Secrets:** Bundle audits verifying zero API keys leak into client cod
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4. Judge Cross-Examination Defense Matrix (Technical Q&A)

Q1: "What if the clinician has a cold or the ICU is noisy?"

> **Answer:** *"WeSpeaker evaluates acoustic consistency and voiced speech duration (≥ 1.2s). If a doctor is hoarse, MEDIQ fails closed—it will never guess on poor audio. Clinicians use our instant touchless fallback (**Demo Quick-Sim** or **Typed commands**) which have identical navigation privileges."*

Q2: "Why dual-model consensus instead of one model?"

> **Answer:** *"Single-model bias is dangerous in clinical medicine. Asking the same model to audit its own prescription draft introduces confirmation bias. We pair **Hermes 3** (DigitalOcean) for drafting with **Google Gemini** for adversarial safety auditing, backed by deterministic offline rules."*

Q3: "How do you handle patient data privacy (HIPAA / PIPEDA)?"

> **Answer:** *"Zero patient data leaves the workstation. All records are synthetic. Audio is converted into mathematical 256-D embeddings in browser WASM and garbage-collected immediately. Only anonymized clinical notes reach cloud AI for drafting."*

Q4: "Why browser MediaPipe instead of dedicated hardware like Leap Motion?"

> **Answer:** *"Deployability. Specialized depth cameras cost hundreds of dollars and require proprietary drivers hospital IT won't approve. Browser WASM turns every existing hospital laptop and cart webcam into an instant touchless workstation."*

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